NFL Third Down Conversions Predictive Analysis

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1 Introduction

The National Football League (NFL) is the professional American football league formed since the 1920's. It was originally founded as the American Professional Football Association (APFA) and later called the NFL two years later ([3]). Today, the league consists of 32 football teams divided into two conferences where each team plays regular 17 regular-season games over an 18-week schedule, including one bye week. The two conferences: the American Football Conference (AFC) and the National Football Conference (NFC) contain the 4 divisions (East, North, South, and West) where 4 teams belong to each division.

In American football, there are many crucial moments during a game, but for an offense, the third-down play is often the last opportunity to continue a drive. Failure to convert a third down results in the offensive team to either punt the ball giving it to the defensive team or risk a possibility of not converting on fourth-down. This study focuses on predicting third-down conversion success using logistic regression on NFL play-by-play data from the 2009 to 2018 seasons. In this paper, I will describe the dataset and provide explanation of the variables selection. Afterwards I will identify the best-fitting logistic regression model, provide real-world interpretations of the predictor variables and characterizing the predictive accuracy of my best model.

2 Data

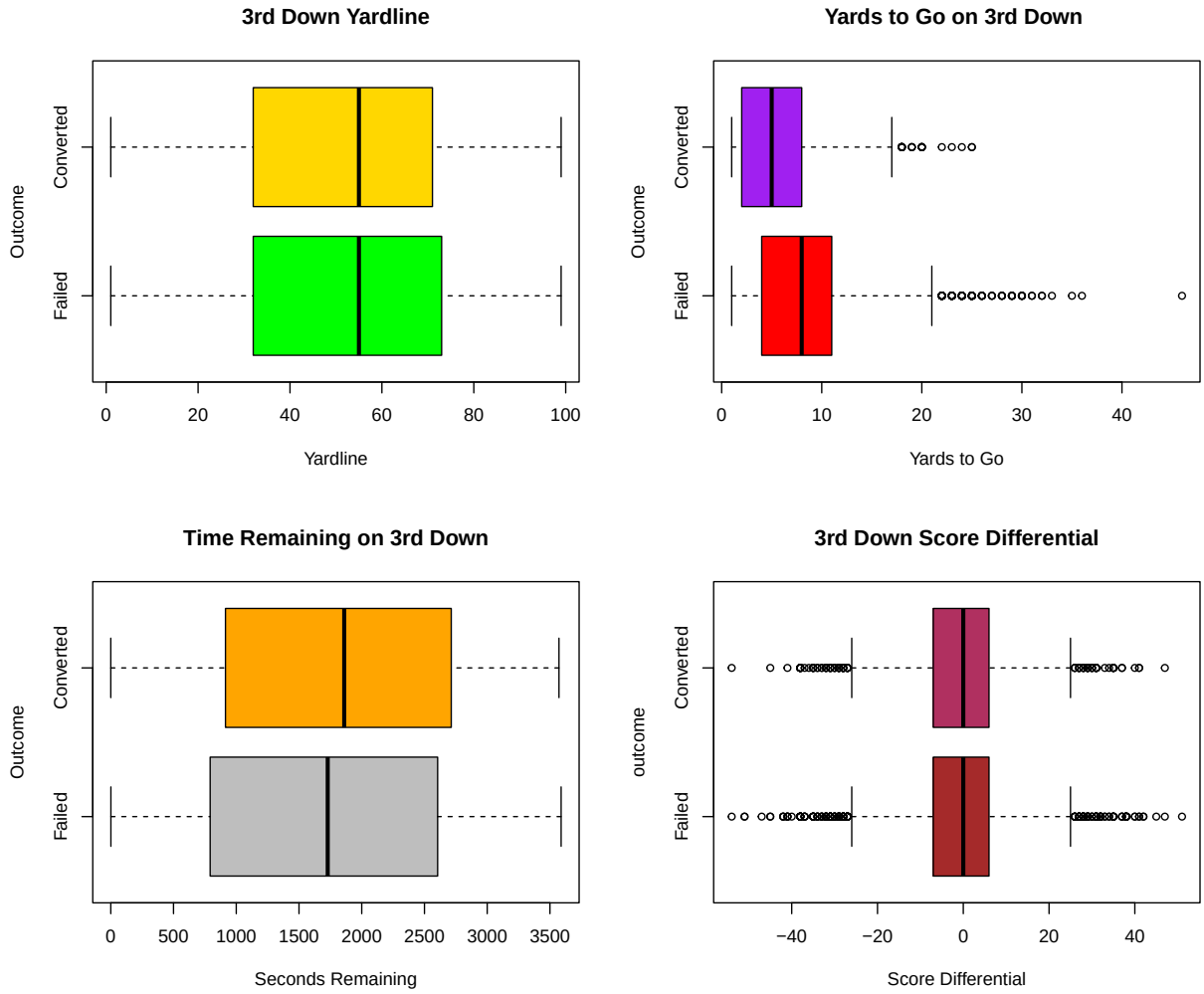
I used the "Detailed NFL Play-by-Play Data 2009-2018" dataset from Kaggle, complied by Max Horowitz, Sam Ventura, and Ron Yurko ([1]). This dataset consists of 449,000 rows and 255 columns of data where each row consists of a play containing information of the game situation, the players involved and the play results. Since the data was too large to analyze, using Microsoft Excel filters I had to clean the dataset. For simplicity, I only accounted for the year 2014, which was the year when I first had interest in football. Afterwards I filtered the data for the year 2014, I further accounted for only third downs by changing the "down" column to 3 which signifies a third-down. Afterwards, I searched for "N/A" entries and removed them to assure accurate results for my binary variable Y which is whether a NFL team converts on third down.

$$Y = \begin{cases} 0 & \text{Failed to Convert on 3rd down} \\ 1 & \text{Convert on 3rd down} \end{cases} \tag{1}$$

I would end up having 7487 third down plays in my NFL 2014 dataset where I would choose my predictors variables x_i. For my 8 chosen predictor variables, I chose 4 quantitative variables: "Yardline100" (Yard Line), "ydstogo" (Yards to Go), "GameSecondsRemaining" (Game Seconds Remaining), and "ScoreDifferential" (Score Differential) with 4 categorical variables: "qtr" (Quarter), "goaltogo" (Goal to Go), "shotgun" (Shotgun), and "nohuddle" (No Huddle).

Table 1: Summary Statistics for Quantitative Variables

Quantitative Variable	**Min**	**1st Qu.**	**Median**	**3rd Qu.**	**Max**	**Mean**	**SD**
Yard Line	1.00	32.00	55.00	72.00	99.00	51.38	24.83
Yards to Go	1.00	3.00	6.00	10.00	46.00	7.22	5.04
Game Seconds Remaining	0	844	1804	2646	3589	1745	1042.80
Score Differential	−54.00	−7.00	0.00	6.00	51.00	−0.95	11.42



For the quantitative (numerical) variables, each distribution is shown above. The Yard Line represents the current yard line and how long of a distance is needed to score a touchdown. Its distribution is skewed to the left where most of the third down plays most commonly happen around the 51 yard line. For Yards to Go, it has a mean of 7.216. This implies meaning the most common yards to convert is about 8 yards. There are some outliers higher yardage in order to convert on third down but that comes with penalties after the team is set and plays occur. For the Game Seconds Remaining, most of the third down plays occur in the 2nd quarter (2,700 seconds) and near the start of the 4th quarter at the ten minute marker (600 seconds). As for Score Differential, when filtering it for only third downs, there

were many outliers with a mean of approximately -1. This distribution looks symmetric where there are situations in the NFL where its complete blowout, one team dominating the other.

For my selected qualitative (categorical) variables, they each have their own indicators/-classifications. The most notable would be the "qtr" variable which represents the game quarter. There are five game quarters: four regular quarters (Q1, Q2, Q3, and Q4) and one overtime (OT), where the overtime only happens if a tie occurs at the end of the last quarter. For the other three variables, they are indicator variables on whether their conditions are met. Goal to Go infers whether the offensive to is near 10 yards to the opponent goal line, Shotgun is determine whether the offensive quarterback (QB) lines about 5 yards back from the center and No Huddle is the game strategy where a team doesn't meet up and with the quarterback to hear the play. For their defining characteristics, they are listed below.

Table 2: Categorical Variable Characteristics

Variable	**Definition**
Qtr	1 = 1st Quarter
	2 = 2nd Quarter
	3 = 3rd Quarter
	4 = 4th Quarter
	5 = Overtime (OT)
Goal to Go	0 = $>$ 10 yards to goal line
	1 = $\leq$ 10 yards to goal line
Shotgun	0 = QB under center
	1 = QB 5 yards back
No Huddle	0 = regular huddle
	1 = no huddle

Table 3: Third Down Conversions by Categorical Variables

Table 4: **Goal to Go**

Goal	Failed	Convert
0	4493	2593
1	266	135

Table 5: **No Huddle**

Huddle	Failed	Convert
0	4332	2426
1	427	302

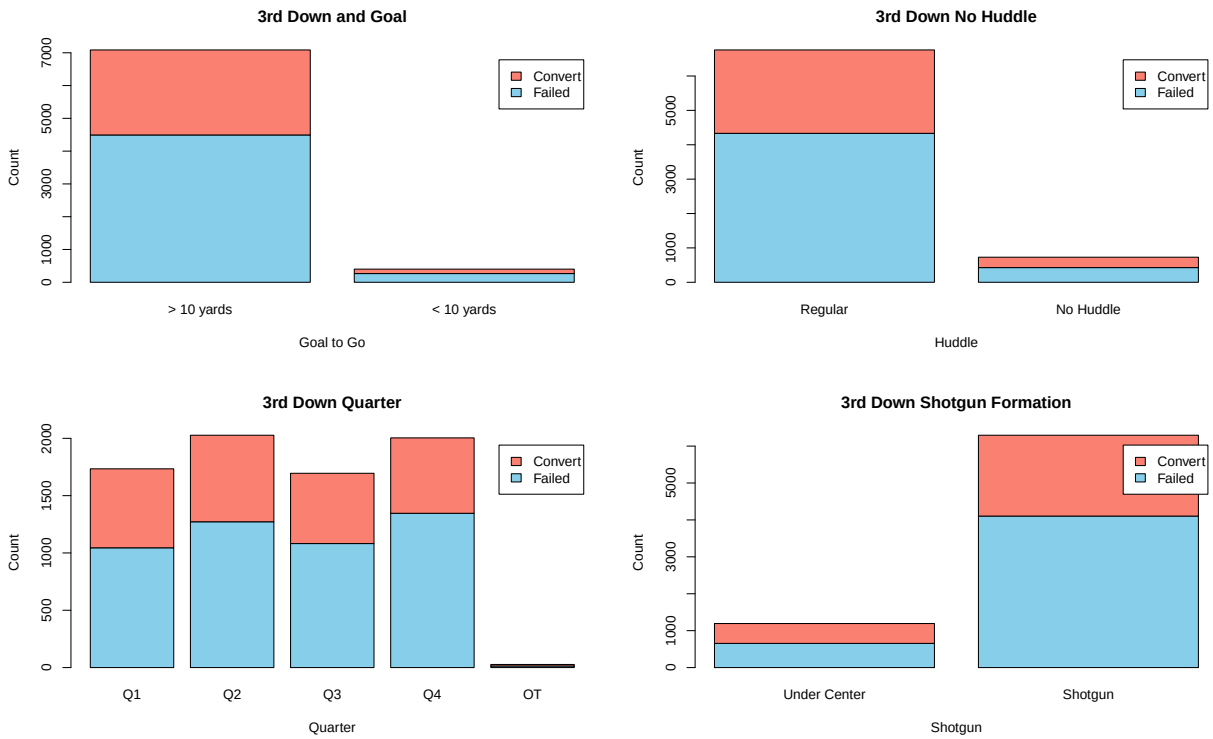
Table 6: **Quarter**

Qtr	Failed	Convert
1	1045	689
2	1272	755
3	1082	613
4	1346	658
5	14	13

Table 7: **Shotgun Formation**

Shotgun	Failed	Convert
0	655	538
1	4104	2190

As for their distributions, bar plots are given along with a color coded difference between them to tell the difference in their proportions. Red for convert and blue for failed. Objectively, each distribution consists of 3rd down conversions that failed. For Third and Goal plot (graph (2,1)), that data consists of general third downs that are more than 10 yards from the first and goal. For No Huddle, many football teams general do a huddle formation, this would make sense since you don't send people down the field by pushing on the same play over again unless its near the end of the game.



As for Quarter, there's not many OT third downs because its "sudden death" were each team gets at least a few tries to break a tie. Its worth noting though that for the second half of each half (Q2 and Q4), there are more third down plays than their counterparts since each team are pressured at the end to score to end the game with a higher score either before the first half ends or the end of the game. For Shotgun, teams are surprisingly form more shotgun formations than under center. Each formation would either be pass or rush and it ultimately depends on the play. Its worth noting that the distributions for the QB being under center is almost the same when comparing it to the no huddle formation formation.

Table 8: 3rd Down Conversions by Quarter

Quarter	**Failed (Y=0)**	**Converted (Y=1)**
Q1	0.603	0.397
Q2	0.628	0.372
Q3	0.638	0.362
Q4	0.672	0.328
OT	0.519	0.481

Table 9: 3rd Down Conversions by Situation

Variable / Category	**Failed (Y=0)**	**Converted (Y=1)**
Goal to Go > 10 yards (0)	0.634	0.366
Goal to Go ≤ 10 yards (1)	0.663	0.337
No Huddle = 0 (Regular)	0.641	0.359
No Huddle = 1 (No Huddle)	0.586	0.414
Shotgun = 0 (Under Center)	0.549	0.451
Shotgun = 1 (Shotgun)	0.652	0.348

3 Model Selection

After defining my binary variable Y as whether a NFL team successfully converts on third-down, a logistic regression model is fitted using eight predictor variables via the `glm()` function in R. Prior to setting up the initial model, some training and testing of the data is required. This is so I can relate the predictor variables to my prediction of third down conversions. It also helps in terms of learning the behaviors of each predictors where a portion of the sample size is taken out of the cleaned dataset of $7,487$ NFL plays. For this dateset I removed 25% of the 2014 NFL plays in my training and analysis of the data. Afterwards, I created the initial model shown below with the glm() function in Rstudio with the family setting to `binomial`.

$$\eta_i = \beta_0 + \beta_1 x_{i,\text{YL}} + \beta_2 x_{i,\text{YTG}} + \beta_3 x_{i,\text{GSR}} + \beta_4 x_{i,\text{SD}} + \beta_5 x_{i,\text{GTG}} + \beta_6 x_{i,\text{NH}} + \beta_7 x_{i,\text{QTR}} + \beta_8 x_{i,\text{SG}},$$

where

- $\eta_i = \log\left(\dfrac{\Pr(Y_i = 1)}{1 - \Pr(Y_i = 1)}\right)$ is the log-odds that play i results in a successful third-down conversion.
- $x_{i,\text{YL}}$ is the yard line (distance from the opponent's end zone) for play i.
- $x_{i,\text{YTG}}$ is the number of yards required to convert on play i.
- $x_{i,\text{GSR}}$ is the number of game seconds remaining at play i.
- $x_{i,\text{SD}}$ is the score differential at the start of play i.
- $x_{i,\text{GTG}}$ is an indicator variable for a goal-to-go situation.
- $x_{i,\text{NH}}$ is an indicator variable for a no-huddle offense.
- $x_{i,\text{QTR}}$ is the quarter of the game.
- $x_{i,\text{SG}}$ is an indicator variable for a shotgun formation.

Using these variables the following results are shown for the initial model on the next page where most of the variables aren't significant (next page). From the initial model, only the first two predictor variables (Yard Line and Yards to Go) are significant. There is

also the null and residual deviance given along with the Akaike Information Criterion (AIC) which are shown to see how the model fits the data. For an initial model, it a good start but there is need of improvements. The goal of this is analysis is to try and predict third down conversions with my predictor variables but also to improve the model. Since Yard Line and the Yards to Go variables are significant, they are power predictors in terms of this initial model. I would add some nonlinear (quadratic x_i^2 and cubic x_i^3) terms for the quantitative variables and interaction terms for each predictor variable in order to improve the model's AIC. This is so that it can fit the data the better and achieve a lower AIC value which is better in terms of this analysis.

Table 10: Initial Logistic Regression Model

Predictor	**Estimate**	**Std. Error**	**z value**	**Pr(>|z|)**
(Intercept)	0.6073	0.4404	1.379	0.1679
yardline_100	**0.00422**	**0.00135**	**3.118**	**0.00182**
ydstogo	**-0.1590**	**0.00779**	**-20.416**	$< 2e^{-16}$
game_seconds_remaining	-0.0000419	0.000108	-0.389	0.6971
score_differential	0.00115	0.00263	0.437	0.6623
goal_to_go	-0.2097	0.1460	-1.436	0.1510
no_huddle	0.0832	0.0962	0.864	0.3874
qtr	-0.1125	0.0995	-1.130	0.2586
shotgun	0.0500	0.0825	0.606	0.5445

Model	Null Deviance	Residual Deviance	DF	AIC
Initial	7376.6	6786.2	5607	6804.2

After doing some trial and error for each term, I added quadratic, cubic, and interaction terms for the following predictors while providing the Null and Residual Deviances and AIC values (shown next page).

NonLinear and Interaction Terms Added:

- Quadratic terms: Yard Line, Yards to Go, Game Seconds Remaining
- Cubic terms: Yard Line, Yards to Go
- Interaction terms:
  - Yard Line x (Yards to Go, Shotgun)
  - Yards to Go x (Game Seconds Remaining, No Huddle, Shotgun)
  - Game Seconds Remaining x (qtr, Shotgun)
  - Score Differential x (qtr, Shotgun)
  - qtr x Shotgun

As for the interaction terms, it would make sense that there could be some interaction influencing each predictor. An example would be for the effect of the yard line. The probability of converting on third down may depend on the yards to go or the play formation. To

continue with the yards to go, the game time and the formation (Huddling and Shotgun) are based on the conditions each team are facing. Game Seconds Remaining may interact with the game quarter or team formation and the same for the Score Differential which influences decision-making depending on the quarter or play formation. After adding these terms I get following values for the deviances and AIC.

Table 11: Logistic Regression Model Deviances and AIC Post Adding Terms

Null Deviance	**Residual Deviance**	**AIC**
7376.6	6119.0	6767

The AIC has decreased slightly with the additional terms but in order to improve the model, I conducted the Hosmer-Lemeshow (HL) test. This is another test in order to achieve the goodness of fit for logistic regression. Usually a lower p-value leads to better results in any statistical test but for the HL test, a higher p-value indicates a better fit of the model. Looking at the initial p-value of the model with the additional terms the p-value from the HL test is 0.64416. Using the HL test multiple times, I systematically looked at the nonsignificant nonlinear and interaction terms for my current model and remove them if the p-value from the HL test increases. After removing the following terms I ended up with an p-value from the HL test of 0.99825.

NonLinear and Interaction Terms Removed:

- Score Differential x Shotgun
- I($yardline100^3$)

Initial p-value = 0.64416; Final p-value = 0.99825

From there I fitted the model and got the coefficients values for my final model shown on the next page. My final model still has some non-significant variables such as the nonlinear terms of the yard line and yards to go as well as interaction terms for between the yard to go and the no huddle but removing them without checking the HL test would worsen the goodness of fit of the logistic model. Also with an AIC of 6768.2, it's lower by 1 point from the previous model but a better fit. For that I would choose this model for my logistic regression analysis on third down conversions.

Table 12: Final Logistic Regression Model Output

Null Deviance	**Residual Deviance**	**AIC**
7376.6	6724.2	6768.2

Table 13: Final Logistic Regression Model

Predictor	**Estimate**	**Std. Error**	**z value**	**Pr(> |z|)**
Predictor	Estimate	Std. Error	z value	Pr(> |z|)
(Intercept)	**-3.634**	**1.342**	**-2.707**	**0.00679**
yardline_100	0.01159	0.00633	1.831	0.06710
I(yardline_100^2)	-0.0001094	0.0000630	-1.736	0.08258
ydstogo	**-0.2545**	**0.05225**	**-4.870**	$1.11e^{-06}$
I(ydstogo^3)	-0.0001325	0.0001812	-0.731	0.46465
game_seconds_remaining	**0.002680**	**0.0007011**	**3.823**	**0.00013**
I(game_seconds_remaining^2)	$-3.835e^{-07}$	$1.112e^{-07}$	**-3.449**	**0.00056**
score_differential	**0.02008**	**0.009712**	**2.068**	**0.03868**
goal_to_go	-0.08102	0.1725	-0.470	0.63859
no_huddle	-0.08234	0.1674	-0.492	0.62276
qtr	**0.8922**	**0.3090**	**2.887**	**0.00389**
shotgun	1.769	1.211	1.460	0.14418
yardline_100:ydstogo	-0.0002714	0.0003507	-0.774	0.43899
yardline_100:shotgun	0.006236	0.003426	1.821	0.06867
ydstogo:game_seconds_remaining	-0.00001358	0.000007538	-1.801	0.07167
ydstogo:no_huddle	0.02657	0.02336	1.137	0.25547
ydstogo:shotgun	**0.1060**	**0.02920**	**3.631**	**0.00028**
game_seconds_remaining:qtr	**-0.0003473**	**0.0001126**	**-3.084**	**0.00204**
game_seconds_remaining:shotgun	-0.0005790	0.0002984	-1.940	0.05238 .
score_differential:qtr	-0.005582	0.002988	-1.868	0.06174 .
qtr:shotgun	**-0.5438**	**0.2754**	**-1.975**	**0.04829**

4 Results

In this section, I will be analyzing the coefficients and interpret their estimations from my final model. I will also characterize the predictive accuracy in this section. For the odds ratios of each term I selected a Δx in order to give real world interpretations. Each Odds Ratio (OR) is calculated as follows:

$$OR = e^{\Delta x * \beta_i} \quad \text{for i} = 1, 2, ..., n \quad \text{coefficients}$$

For each predictor, assuming all other variables are held constant:

- Yard Line OR: For each 10 yard increase towards the opponent's end zone ($\Delta x = 10$), the odds of converting on third down increases by 12.3%
- Yards to Go OR: For each 5 additional 5 yards gain ($\Delta x = 5$), the odds of converting on third down decreases by 72%
- Game Seconds Remaining OR: For each additional 60 seconds remaining in the game ($\Delta x = 60$) , the odds of converting on third down increase by 17.4%
- Score Differential OR = For each additional point ($\Delta x = 1$) , the odds of converting on third down increases by 2.0%

- Goal to Go OR = Going from far away to near the goal line ($\Delta x = 1$) , the odds of converting on third down decreases by 7.8%
- Quarter OR = Going to the next quarter ($\Delta x = 1$), the odds of converting on third down increases by 144%
- No Huddle OR = Having a huddle then not having a huddle ($\Delta x = 1$), decreases to the odds of converting on third down by 7.9%
- Shotgun OR = Going from not having a shotgun formation to having it ($\Delta x = 1$) increases the odds of converting on third down by 486%.

These predictor variables' interpretation would make sense. In terms every additional Δx yards or going from each categorical section, the odds of converting on third down would generally depend for each addition set of yards or preplay formation. For Yard Line, Game Seconds Remaining, Score Differential, Quarter and Shotgun formation, they each increases the chances of converting on third down. As for the other variables (Yards to Go, Goal to Go and No Huddle), they each decreases the chance of converting. This would make sense since as you get more sets of downs you tend to not always need a third down if a drive is successful but its different when getting close to the goal line and having no huddle. For doing confidence intervals (CI's) it would be the similar circumstances by multiply them with their respective Δx's. For more on their outputs they are attached in the appendices output (page 13).

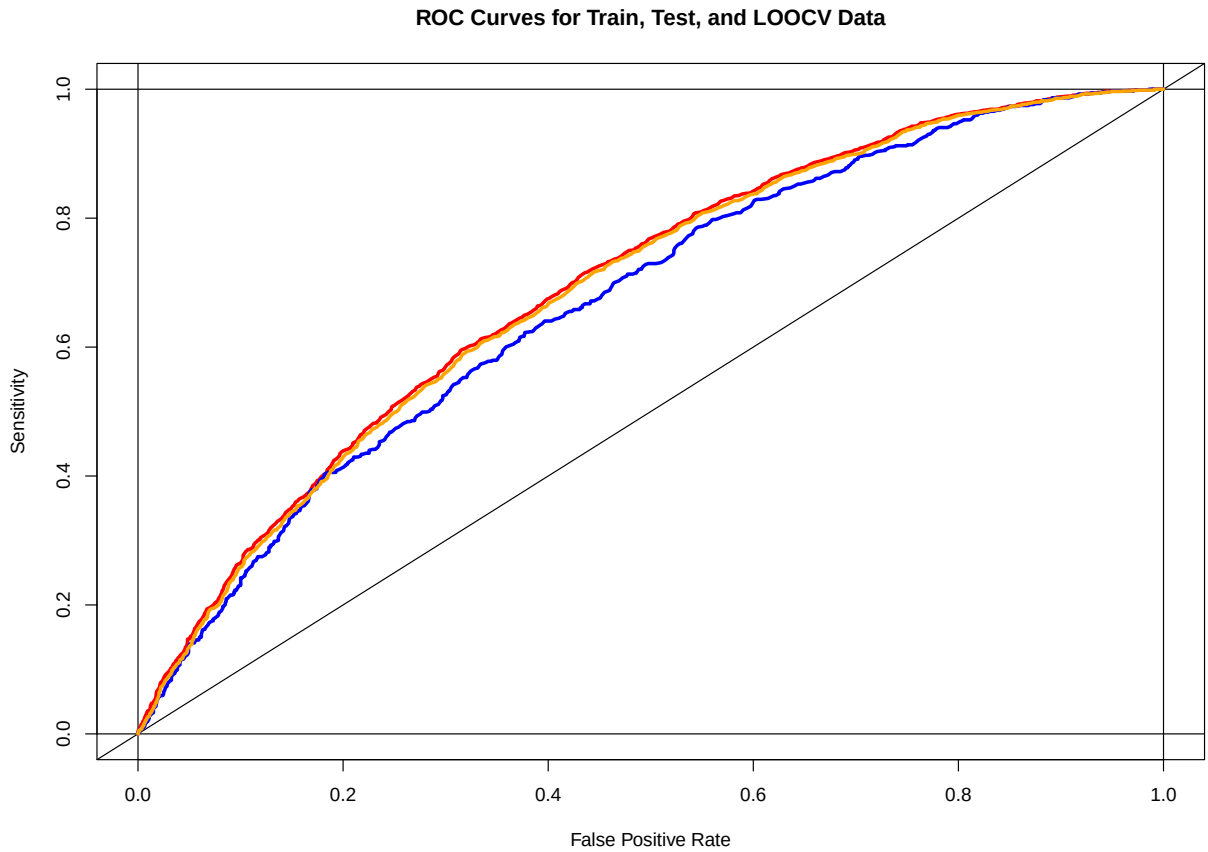
Table 14: Predictive Accuracies for Logistic Regression Model

Data	**Accuracy**
Training	0.6681
Test	0.6660
LOOCV	0.6654

For the model's overall accuracies of computed from their respected data, they are shown above. Three accuracies were calculated to determine the overall predictive ability from the final model to predict third down conversions. In terms of the three outputs, three accuracies were computed from the training and testing data as well in the leave-one-out cross-validated (LOOCV) procedure. The LOOCV is conducted on the entire dataset and since included nonlinear and interaction terms, this would ensure that the model is not over fitted. LOOCV takes one same out and does a prediction for the that specific dataset and does it again n times. Based on the accuracies on between the training, testing and LOOCV data, the numbers are roughly the same in terms of overall predictive accuracies which indicate the model having at least 0.66 predictive power on third down conversions.

Receiving Operating Characteristic (ROC) curves were constructed for each of the datasets. These curves evaluated the model's performance with the three types of data. Based on the output, there was a consistent overlap despite adding new nonlinear terms in the model. When comparing the training and testing data, the testing data was the one below the training and the LOOCV data. Adding new terms for the model would suggest I would get a better ROC Curve which would limit the False Positive Rate and increase the ability to

determine third down conversions. I would say that adding more terms may have overfitting the model but since the accuracies were consistent with each other, there isn't enough information in order to predict third down conversions. Since I set a cutoff of $\tau = 0.5$, which is where it either converts or does not when above or below the parameter, it would fit close to the model's predictions.



5 Summary

In this report, a logistic regression model was developed to predict third-down conversion success in the 2014 NFL season using eight predictors, including four quantitative variables (yard line, yards to go, game seconds remaining, score differential) and four categorical variables (Quarter, Goal to Go, Shotgun, No Huddle). The initial model indicating the yard line and yards to go were significant predicts and addition nonlinear terms and quadratic terms improved the model's fit. When conducting the Hosmer-Lemeshow test, I achieved a goodness of fit of my data which has a p-value of 0.99825. This model is better in interpreting the odds with a slightly improved AIC (6768.2) compared to the initial model's AIC. Interpreting the odds ratios of each predictor variable reveals a higher probability of convergence for higher yard lines and score differentials as well as for game time remaining, later quarters, and shotgun formation. For the variables yards to go, goal-to-go and no huddle offense tend to decrease the odds of converting.

For the predictive accuracy of the three data, they were overall consistent with each other. The training data, testing data and LOOCV each had an overall accuracy of about 66%. This would be supported through an ROC curve analysis. The ROC curves were mostly overlapped with one another while the testing data curve is lower than the training and LOOCV data. It may suggest that I may have over fitted but the model captures seems to capture the general trend of third-down conversion outcomes. However when initially testing the data, adding nonlinear terms and interaction terms didn't overall improve the model's accuracy to predict third down conversions. I would say that adding more variables such as the play type, formation or run/pass location would help. This dataset did have a play type but they were only found out after the play commenced. For logistic regression there needs to be a predetermined play action a variable to be included in this model and it doesn't help. A similar NFL data analysis reveals the overall accuracy in third down conversions reveal an accuracy of 0.6560. This was conducted by a Kaggle user who did a similar code but in Python which indicate that my model seems to be accurate in terms of the data ([2]).

Overall I would say that it is hard to predict third down conversions since there are many unknown factors that can happen after and prior to the play action. There is not much to record on except only on the football location, the team formation and the Expected Points Added (EPA) statistics which are post play. Other variables that can improve this analysis would be the weather and terrain as well as training and fitness of the players but they were not given.

References

[1] Max Horowitz. Nflplaybyplay2009to2016 dataset. https://www.kaggle.com/datasets/maxhorowitz/nflplaybyplay2009to2016, 2025. Detailed NFL play-by-play data (2009–2016) from Kaggle; accessed on March 17, 2026.

[2] jamiewelsh2. Nfl data analysis – predicting third down success. https://www.kaggle.com/code/jamiewelsh2/nfl-data-analysis-predicting-third-down-success/notebook, 2025. Kaggle Notebook; accessed on March 17, 2026.

[3] YouTube. Nfl third down analytics (video). https://www.youtube.com/watch?v=ed9346Z4uLk, 2025. YouTube video; accessed on March 17, 2026.

6 Appendices

6.1 Tables

Table 15: 95% Confidence Intervals for Predictor Variables

Predictor	**2.5%**	**97.5%**
Intercept	-6.2743	-1.0067
yardline_100	-0.0007936	0.024024
I(yardline_100$\hat{2}$)	-0.0002332	0.00001381
ydstogo	-0.35967	-0.15540
I(ydstogo$\hat{2}$)	-0.005844	0.01640
I(ydstogo$\hat{3}$)	-0.0005247	0.0001721
game_seconds_remaining	0.001307	0.004057
I(game_seconds_remaining$\hat{2}$)	-6.0168e-07	-1.65697e-07
score_differential	0.001066	0.03915
goal_to_go	-0.42093	0.25564
no_huddle	-0.40834	0.24842
qtr	0.28806	1.50083
shotgun	-0.60537	4.14660
yardline_100:ydstogo	-0.0009556	0.0004198
yardline_100:shotgun	-0.0004741	0.012961
ydstogo:game_seconds_remaining	-2.84073e-05	1.15812e-06
ydstogo:no_huddle	-0.02042	0.07137
ydstogo:shotgun	0.05074	0.16528
game_seconds_remaining:qtr	-0.00056798	-0.00012653
game_seconds_remaining:shotgun	-0.001165	5.553e-06
score_differential:qtr	-0.011446	0.0002705
qtr:shotgun	-1.08515	-0.004665

Table 16: Correlation Matrix of Predictors and Third Down Conversion

	Third Down Conversion	Game Seconds Remaining	Yard Line 100	Yards to Go	Score Diff	Goal To Go	No Huddle	Quarter	Shotgun
Third Down Conversion	1.0000	0.0538	-0.0146	-0.2930	0.0171	-0.0137	0.0341	-0.0485	-0.0783
Game Seconds Remaining	0.0538	1.0000	0.0600	-0.0468	-0.0141	-0.0118	-0.0008	-0.9650	0.0944
Yardline 100	-0.0146	0.0600	1.0000	0.1950	-0.0589	-0.4371	0.0124	-0.0309	0.1147
Yards to Go	-0.2930	-0.0468	0.1950	1.0000	-0.0426	-0.0685	-0.0427	0.0358	0.2619
Score Differential	0.0171	-0.0141	-0.0589	-0.0426	1.0000	-0.0014	-0.0534	0.0203	-0.1389
Goal To Go	-0.0137	-0.0118	-0.4371	-0.0685	-0.0014	1.0000	-0.0261	-0.0005	-0.0553
No Huddle	0.0341	-0.0008	0.0124	-0.0427	-0.0534	-0.0261	1.0000	-0.0013	0.0630
Quarter	-0.0485	-0.9650	-0.0309	0.0358	0.0203	-0.0005	-0.0013	1.0000	-0.0770
Shotgun	-0.0783	0.0944	0.1147	0.2619	-0.1389	-0.0553	0.0630	-0.0770	1.0000

6.2 R Code

```r
library(readr)
NFL_Play_by_Play_3rd_down_2 <- read_csv("Fall 2025/STA 4350/final
   project/dataset searches/Use/NFL Play by Play 3rd down_2.csv")
View(NFL_Play_by_Play_3rd_down_2)

# any N/A values
any(is.na(NFL_Play_by_Play_3rd_down_2))

# rename
NFL_2014 = NFL_Play_by_Play_3rd_down_2

# first rows
head(NFL_2014)

# third down conversion
Y = NFL_2014$third_down_converted
Y

# predictor variables
yardline_100 = NFL_2014$yardline_100
ydstogo = NFL_2014$ydstogo
game_seconds_remaining = NFL_2014$game_seconds_remaining
score_differential = NFL_2014$score_differential
qtr = NFL_2014$qtr
goal_to_go = NFL_2014$goal_to_go
shotgun = NFL_2014$shotgun
no_huddle = NFL_2014$no_huddle

# quantitative
game_seconds_remaining;
yardline_100;
ydstogo;
score_differential

```

```r
# qualitative
goal_to_go;
no_huddle;
qtr;
shotgun;

qtr = as.factor(qtr)

# size of sample
length(Y)
# 7487

#1) describe data

#quantitative: five number summaries
summary(yardline_100)
summary(ydstogo)
summary(game_seconds_remaining)
summary(score_differential)
sd(yardline_100); sd(ydstogo); sd(game_seconds_remaining); sd(score_differential)

par(mfrow=c(2,2))
# Boxplot: yardline_100 vs third_down_converted
boxplot(yardline_100 ~ factor(Y, labels = c("Failed", "Converted")),
        horizontal = TRUE,
        main = "3rd Down Yardline",
        xlab = "Yardline",
        ylab = "Outcome",
        col = c("green","gold"))

# Boxplot: ydstogo vs third_down_converted
boxplot(ydstogo ~ factor(Y, labels = c("Failed", "Converted")),
        horizontal = TRUE,
        main = "Yards to Go on 3rd Down ",
        xlab = "Yards to Go",
        ylab = "Outcome",
        col = c("red","purple"))

# Boxplot: game_seconds_remaining vs third_down_converted
boxplot(game_seconds_remaining ~ factor(Y, labels = c("Failed", "Converted")),
        horizontal = TRUE,
        main = "Time Remaining on 3rd Down",
        xlab = "Seconds Remaining",
        ylab = "Outcome",
        col = c("grey","orange"))
```

```r

# Boxplot: score_differential vs third_down_converted
boxplot(score_differential ~ factor(Y, labels = c("Failed", "Converted")),
        horizontal = TRUE,
        main = "3rd Down Score Differential",
        xlab = "Score Differential",
        ylab = "outcome",
        col = c("brown","maroon"))

par(mfrow=c(1,1))


# categorical=======
table(goal_to_go,Y)
table(no_huddle,Y)
table(qtr,Y)
table(shotgun,Y)

prop.table(table(goal_to_go,Y),margin=1)
prop.table(table(no_huddle,Y),margin=1)
prop.table(table(qtr,Y),margin=1)
prop.table(table(shotgun,Y),margin=1)

prop.test(table(goal_to_go,Y),correct=FALSE)
prop.test(table(no_huddle,Y),correct=FALSE)
prop.test(table(qtr,Y),correct=FALSE)
prop.test(table(shotgun,Y),correct=FALSE)

# categorical plots
par(mfrow=c(2,2))

# Barplot for goal to goal
prop_goal <- table(NFL_2014$goal_to_go, Y)

barplot(t(prop_goal),
        col = c("skyblue", "salmon"),
        legend.text = c("Failed","Convert"),
        main = "3rd Down and Goal",
        xlab = "Goal to Go",
        ylab = "Count",
        names.arg = c("> 10 yards", "< 10 yards"))  # 0 = No, 1 = Yes
  # asked chat help for changing the labels

# Barplot of no_huddle vs Y
prop_huddle <- table(NFL_2014$no_huddle, Y)
```

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barplot(t(prop_huddle),
        col = c("skyblue", "salmon"),
        legend.text = c("Failed","Convert"),
        main = "3rd Down No Huddle",
        xlab = "Huddle",
        ylab = "Count",
        names.arg = c("Regular", "No Huddle"))  # 0 = Regular, 1 = No Huddle
# asked chat help for changing the labels

# Barplot of qtr vs Y
prop_qtr <- table(NFL_2014$qtr, Y)
barplot(t(prop_qtr),
        col = c("skyblue", "salmon"),
        legend.text = c("Failed","Convert"),
        main = "3rd Down Quarter",
        xlab = "Quarter",
        ylab = "Count",
        names.arg = c("Q1", "Q2", "Q3", "Q4", "OT"))
# asked chat help for changing the labels

# Barplot of shotgun vs Y
prop_shotgun <- table(NFL_2014$shotgun, Y)

barplot(t(prop_shotgun),
        col = c("skyblue", "salmon"),
        legend.text = c("Failed","Convert"),
        main = "3rd Down Shotgun Formation",
        xlab = "Shotgun",
        ylab = "Count",
        names.arg = c("Under Center", "Shotgun"))  # 0 = Under Center, 1 = Shotgun
# asked chat help for changing the labels

par(mfrow=c(1,1))

# correlation
# variables for correlation matrix
vars <- c("third_down_conversion",
          "game_seconds_remaining",
          "yardline_100",
          "ydstogo",
          "score_differential",
          "goal_to_go",
          "no_huddle",
          "qtr",
```

```r
          "shotgun")


# corelation matrix
with(NFL_2014, cor(data.frame("Third Down conversion" = Y,
  "Game Seconds Remaining" = game_seconds_remaining,
                              "Yardline 100" = yardline_100,
                              "Yards to Go" = ydstogo,
                              "Score Differential" = score_differential,
                              "Goal To Go" = as.numeric(goal_to_go),
                              "No Huddle" = as.numeric(no_huddle),
                              "Quarter" = as.numeric(qtr),
                              "Shotgun" = as.numeric(shotgun))))
# correlation scatter plot
with(NFL_2014, pairs(data.frame("Third Down conversion" = Y,
                                "Game Seconds Remaining" = game_seconds_remaining,
                                "Yardline 100" = yardline_100,
                                "Yards to Go" = ydstogo,
                                "Score Differential" = score_differential,
                                "Goal To Go" = as.numeric(goal_to_go),
                                "No Huddle" = as.numeric(no_huddle),
                                "Quarter" = as.numeric(qtr),
                                "Shotgun" = as.numeric(shotgun) ) ) )

#=======
# 2) Model selection

# Train/Test model
set.seed(4350)
n_total = nrow(NFL_2014)
n_test = floor(0.25 * n_total) # need a whole number
n_test #1871
test_i = sample(n_total, n_test)

NFL_test = NFL_2014[test_i, ]
NFL_train = NFL_2014[-test_i, ]

#Build the model
mlr1 = glm(third_down_converted ~ yardline_100 + ydstogo + game_seconds_remaining +
             score_differential + goal_to_go + no_huddle + qtr + shotgun,
```

```r
            family = binomial, data = NFL_train)
summary(mlr1)

# trial and error
mlr2 = glm(third_down_converted ~ yardline_100 + I(yardline_100^2) + I(yardline_100^3)
            + ydstogo + game_seconds_remaining +
              score_differential + goal_to_go + no_huddle + qtr + shotgun,
            family = binomial, data = NFL_train)
summary(mlr2)

mlr3 = glm(third_down_converted ~ yardline_100 + I(yardline_100^2) + I(yardline_100^3)
            + ydstogo + I(ydstogo^2) + I(ydstogo^3) + game_seconds_remaining +
              score_differential + goal_to_go + no_huddle + qtr + shotgun,
            family = binomial, data = NFL_train)
summary(mlr3)

mlr4 = glm(third_down_converted ~ yardline_100 + I(yardline_100^2) + I(yardline_100^3)
            + ydstogo + I(ydstogo^2) + I(ydstogo^3) + game_seconds_remaining +
              I(game_seconds_remaining^2) +
              score_differential + goal_to_go + no_huddle + qtr + shotgun,
            family = binomial, data = NFL_train)
summary(mlr4)

mlr5 = glm(third_down_converted ~ yardline_100 + I(yardline_100^2) + I(yardline_100^3)
            + ydstogo + I(ydstogo^2) + I(ydstogo^3) + game_seconds_remaining +
              I(game_seconds_remaining^2) + score_differential
            +  goal_to_go + no_huddle + qtr + shotgun,
            family = binomial, data = NFL_train)
summary(mlr5)

mlr6 = glm(third_down_converted ~ yardline_100 + I(yardline_100^2) + I(yardline_100^3)
            + ydstogo + I(ydstogo^2) + I(ydstogo^3) + game_seconds_remaining +
              I(game_seconds_remaining^2) + score_differential
            +  goal_to_go + no_huddle + qtr + shotgun + yardline_100:ydstogo + yardline_100:shotgun,
```

```
            family = binomial, data = NFL_train)
summary(mlr6)

mlr7 = glm(third_down_converted ~ yardline_100 + I(yardline_100^2) + I(yardline_100^3)
            + ydstogo + I(ydstogo^2) + I(ydstogo^3) + game_seconds_remaining +
              I(game_seconds_remaining^2) + score_differential
            +  goal_to_go + no_huddle + qtr + shotgun + yardline_100:ydstogo
            + yardline_100:shotgun + ydstogo:game_seconds_remaining + ydstogo:no_huddle
            + ydstogo:shotgun,
            family = binomial, data = NFL_train)
summary(mlr7)

mlr8 = glm(third_down_converted ~ yardline_100 + I(yardline_100^2) + I(yardline_100^3)
            + ydstogo + I(ydstogo^2) + I(ydstogo^3) + game_seconds_remaining +
              I(game_seconds_remaining^2) + score_differential
            +  goal_to_go + no_huddle + qtr + shotgun + yardline_100:ydstogo
            + yardline_100:shotgun + ydstogo:game_seconds_remaining + ydstogo:no_huddle
            + ydstogo:shotgun + game_seconds_remaining:qtr + game_seconds_remaining:shotgun,
            family = binomial, data = NFL_train)
summary(mlr8)

mlr9= glm(third_down_converted ~ yardline_100 + I(yardline_100^2) + I(yardline_100^3)
            + ydstogo + I(ydstogo^2) + I(ydstogo^3) + game_seconds_remaining +
              I(game_seconds_remaining^2) + score_differential
            +  goal_to_go + no_huddle + qtr + shotgun + yardline_100:ydstogo
            + yardline_100:shotgun + ydstogo:game_seconds_remaining + ydstogo:no_huddle
            + ydstogo:shotgun + game_seconds_remaining:qtr + game_seconds_remaining:shotgun
           + score_differential:qtr + score_differential:shotgun,
            family = binomial, data = NFL_train)
summary(mlr9)

mlr10= glm(third_down_converted ~ yardline_100 + I(yardline_100^2) + I(yardline_100^3)
```

```
          + ydstogo + I(ydstogo^2) + I(ydstogo^3) + game_seconds_remaining +
            I(game_seconds_remaining^2) + score_differential
          +  goal_to_go + no_huddle + qtr + shotgun + yardline_100:ydstogo
          + yardline_100:shotgun + ydstogo:game_seconds_remaining + ydstogo:no_huddle
          + ydstogo:shotgun + game_seconds_remaining:qtr + game_seconds_remaining:shotgun
          + score_differential:qtr + score_differential:shotgun + qtr:shotgun,
          family = binomial, data = NFL_train)
summary(mlr10)

# mlr10 is the best; looking at the AIC, now will do by mlrtest
library(glmtoolbox)
hltest(mlr10) # p-value = 0.64416

mlr11= glm(third_down_converted ~ yardline_100 + I(yardline_100^2) + I(yardline_100^3)
           + ydstogo + I(ydstogo^2) + I(ydstogo^3) + game_seconds_remaining +
             I(game_seconds_remaining^2) + score_differential
           +  goal_to_go + no_huddle + qtr + shotgun + yardline_100:ydstogo
           + yardline_100:shotgun + ydstogo:game_seconds_remaining + ydstogo:no_huddle
           + ydstogo:shotgun + game_seconds_remaining:qtr + game_seconds_remaining:shotgun
           + score_differential:qtr + qtr:shotgun,
           family = binomial, data = NFL_train)
summary(mlr11)
hltest(mlr11)

mlr12 = glm(third_down_converted ~ yardline_100 + I(yardline_100^2)
           + ydstogo + I(ydstogo^2) + I(ydstogo^3) + game_seconds_remaining +
             I(game_seconds_remaining^2) + score_differential
           +  goal_to_go + no_huddle + qtr + shotgun + yardline_100:ydstogo
           + yardline_100:shotgun + ydstogo:game_seconds_remaining + ydstogo:no_huddle
           + ydstogo:shotgun + game_seconds_remaining:qtr + game_seconds_remaining:shotgun
           + score_differential:qtr + qtr:shotgun,
           family = binomial, data = NFL_train)
summary(mlr12)
```

```r
hltest(mlr12)

mlr13 = glm(third_down_converted ~ yardline_100 + I(yardline_100^2)
            + ydstogo + I(ydstogo^2) + I(ydstogo^3) + game_seconds_remaining +
              I(game_seconds_remaining^2) + score_differential
            +  goal_to_go + no_huddle + qtr + shotgun + yardline_100:ydstogo
            + yardline_100:shotgun + ydstogo:game_seconds_remaining + ydstogo:no_huddle
            + ydstogo:shotgun + game_seconds_remaining:qtr + game_seconds_remaining:shotgun
            + score_differential:qtr + qtr:shotgun,
            family = binomial, data = NFL_train)
summary(mlr13)
hltest(mlr13)
# p-value =  0.99825

# Model Selection; compare AIC
AIC(mlr1, mlr2, mlr3, mlr4, mlr5,
    mlr6, mlr7, mlr8, mlr9, mlr10,
    mlr11, mlr12, mlr13)

AIC(mlr13)
# 6768.198
# mlr13 due to AIC and hltest

#3) Results

# confidence interval
confint(mlr13)

# likelihood test
anova(mlr1, mlr13, test = "LRT")

# point estimate approximations
# coefficients of mlr14
length(mlr13$coefficients)
mlr13$coefficients
B0 = mlr13$coefficients[1]
B1 = mlr13$coefficients[2]
B2 = mlr13$coefficients[3]
B3 = mlr13$coefficients[4]
B4 = mlr13$coefficients[5]
B5 = mlr13$coefficients[6]
B6 = mlr13$coefficients[7]
B7 = mlr13$coefficients[8]
```

```
B8 = mlr13$coefficients[9]
B9 = mlr13$coefficients[10]
B10 = mlr13$coefficients[11]
B11 = mlr13$coefficients[12]
B12 = mlr13$coefficients[13]
B13 = mlr13$coefficients[14]
B14 = mlr13$coefficients[15]
B15 = mlr13$coefficients[16]
B16 = mlr13$coefficients[17]
B17 = mlr13$coefficients[18]
B18 = mlr13$coefficients[19]
B19 = mlr13$coefficients[20]
B20 = mlr13$coefficients[21]
B21 = mlr13$coefficients[22]

B0; B1; B2; B3; B4;
B5; B6; B7; B8; B9;
B10; B11; B12; B13; B14;
B15; B16; B17; B18; B19;
B20; B21;

B0_hat_pointtest = exp(1 * B0)
B1_hat_pointest = exp(10 * B1)
B2_hat_pointest = exp(100 * B2)
B3_hat_pointest = exp(5 * B3)
B4_hat_pointest = exp(25 * B4)
B5_hat_pointest = exp(125 * B5)
B6_hat_pointest = exp(60 * B6)
B7_hat_pointest = exp(3600 * B7)
B8_hat_pointest = exp(1 * B8)
B9_hat_pointest = exp(1 * B9)
B10_hat_pointest = exp(1 * B10)
B11_hat_pointest = exp(1 * B11)
B12_hat_pointest = exp(1 * B12)
B13_hat_pointest = exp(50*5 * B13)
B14_hat_pointest = exp(10*1 * B14)
B15_hat_pointest = exp(5*60 * B15)
B16_hat_pointest = exp(5*1 * B16)
B17_hat_pointest = exp(5*1 * B17)
B18_hat_pointest = exp(60*1 * B18)
B19_hat_pointest = exp(60*1 * B19)
B20_hat_pointest = exp(1*1 * B20)
B21_hat_pointest = exp(1*1 * B21)

B1_hat_pointest;
B2_hat_pointest;
B3_hat_pointest;
```

```r
B4_hat_pointest;
B5_hat_pointest;
B6_hat_pointest;
B7_hat_pointest;
B8_hat_pointest;
B9_hat_pointest;
B10_hat_pointest;
B11_hat_pointest;
B12_hat_pointest;
B13_hat_pointest;
B14_hat_pointest;
B15_hat_pointest;
B16_hat_pointest;
B17_hat_pointest;
B18_hat_pointest;
B19_hat_pointest;
B20_hat_pointest;
B21_hat_pointest;

# 4)  ROC curves
make_roc <- function(y,ph,clr="red",new_plot=TRUE,dt=0.001) {

  # sequence of tau values at which to evaluate and plot (fpr(tau), sens(tau)):
  taus <- seq(0,1,dt)

  # create vectors to store the sensitivity and fpr values for each tau value:
  sens <- numeric(length(taus))
  fpr <- numeric(length(taus))

  # calculate the sensitivity and fpr values:
  for (i in seq_along(taus)) {
    sens[i] <- mean(ph[y==1] > taus[i])
    fpr[i] <- mean(ph[y==0] > taus[i])
  }

  # if the user wants a new plotting window, make one:
  if (new_plot)
  {
    plot(NA,NA,type="n",xlim=c(0,1),ylim=c(0,1),
         xlab="False Positive Rate",ylab="Sensitivity")
    abline(h=c(0,1),v=c(0,1),a=0,b=1)
  }

  # draw the ROC curve:
  lines(fpr,sens,col=clr,lwd=3)
```

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}

# train data
t = 0.5

phat_train <- predict(mlr13, newdata = NFL_train, type="response")

# train ROC curve
make_roc(NFL_train$third_down_converted, phat_train) # ROC CURVE
yhat_train <- as.numeric(phat_train > t)

train_acc <- mean(yhat_train == NFL_train$third_down_converted)
train_acc
# 0.6680912

# Test
phat_test <- predict(mlr13, newdata=NFL_test, type="response")
yhat_test <- as.numeric(phat_test > t)

test_acc <- mean(yhat_test == NFL_test$third_down_converted)
test_acc
# 0.665954

# Test ROC
make_roc(NFL_test$third_down_converted, phat_test, clr="blue", new_plot = FALSE)

# LOOCV plot
# make a vector to store the predicted probabilities for 7487 observations-
n_train <- nrow(NFL_train)
pi_hat_cv <- numeric(n_train)

system.time(
  for(i in 1:n_train){

    # fit the model excluding observation i:
    mlr13 = glm(third_down_converted ~ yardline_100 + I(yardline_100^2)
                + ydstogo + I(ydstogo^2) + I(ydstogo^3) + game_seconds_remaining +
                I(game_seconds_remaining^2) + score_differential
                + goal_to_go + no_huddle + qtr + shotgun + yardline_100:ydstogo
                + yardline_100:shotgun + ydstogo:game_seconds_remaining + ydstogo:no_huddle
```

```r
                 + ydstogo:shotgun + game_seconds_remaining:qtr + game_seconds_remaining:shotgun
                 + score_differential:qtr + qtr:shotgun,
                 family = binomial, data = NFL_train, subset = -i)

    # predict on observation i:
    pi_hat_cv[i] <- predict(mlr13, newdata = NFL_train[i, ], type="response")
  }
)

# LOOCV
loocv_acc <- mean(as.numeric(pi_hat_cv > t) == NFL_train$third_down_converted)
loocv_acc
# 0.6654202

make_roc(NFL_train$third_down_converted, pi_hat_cv, clr="orange", new_plot=FALSE)

#=======

# Training, Test, LOOCV Accuracies
train_acc; test_acc; loocv_acc
# 0.6680912; 0.665954; 0.6654202

# ROC PLOTS SUMMARIZE

  # ROC FOR TRAIN DATA
  make_roc(NFL_train$third_down_converted, phat_train)

  # ROC FOR TEST DATA
  make_roc(NFL_test$third_down_converted, phat_test, clr="blue", new_plot = FALSE)

  # ROC FOR LOOCV
  make_roc(NFL_train$third_down_converted, pi_hat_cv, clr="orange", new_plot=FALSE)

  title(main = "ROC Curves for Train, Test, and LOOCV Data")
  legend("bottomright", inset=0.05,
         legend = c("Train Data", "Test Data", "LOOCV Data"),
         text.col = c("red", "blue", "orange"))
```